

NETWORKS

A network is created when two or more computers are connected together. Using a network, a computer can communicate with others and share resources such as hardware, software and data. It does this by sending data in a packet, which is a unit of data packaged together so that it can travel across a network.

In order to connect to a network, computers need to fulfil certain requirements so that they all operate using standard protocols. Protocols are Rules that allow the exchange and transmission of data between devices.

NETWORK OPERATING SYSTEMS

A network operating system allows a computer to communicate on a network. It provides additional functionality to a stand-alone operating system, including:

- Passing usernames and passwords to a server for checking when a user logs in.
 - Separating user accounts and ensuring that users cannot access each other's files.
 - Providing access to network storage and shared resources such as networked printers.
-

HOW DEVICES ARE IDENTIFIED ON A NETWORK

There are three methods used to identify devices on a network:

1. Internet Protocol (IP)
2. MAC address
3. Device name

Internet Protocol (IP)

An IP address is a unique address that networked devices use to send data to each other. Each piece of data that is sent across a network carries the IP address of the destination, so that each device in the network knows where to send it.

- **IPv4:** Uses four groups of up to three numbers separated by full stops (for example, `192.168.1.1`). IPv4 can store over 4 billion addresses.
- **IPv6:** Uses eight groups of four hexadecimal numbers separated by colons (for example, `2001:0db8:0000:0042:0000:8a2e:0370:7334`).
- **Allocation:** IP addresses can either be assigned by a network administrator or allocated dynamically by a server running Dynamic Host Configuration Protocol (DHCP).

- **Hexadecimal:** A base-16 number system that uses the numbers 0–9 and the letters A–F.
- **Network administrator:** A person who manages an organisation's network.
- **Dynamically:** In a way that is open to change.
- **Dynamic Host Configuration Protocol (DHCP) server:** A networked computer that automatically assigns an IP address to other computers when they join the network.

MAC Address

Unlike IP addresses, which can be dynamically allocated by users or servers, media access control (MAC) addresses are universally unique identifiers given to the network interface card (NIC). MAC addresses are used in devices connected to local area networks (LANs) using Ethernet, Bluetooth or Wi-Fi. MAC addresses are assigned by the NIC manufacturer and are generally considered to be fixed addresses.

A MAC address is usually made up of six pairs of characters. The first three pairs identify the manufacturer and the remaining pairs are assigned by the manufacturer to uniquely identify the device.

Uses of MAC Addresses

- **Restricting or allowing access to a network:** MAC address filtering checks the MAC address of devices attempting to gain access to a network and only grants access to devices with specified MAC addresses.
- **Identifying a device on a network:** Some Wi-Fi hotspots only provide free access for a certain length of time, and they identify a device using its MAC address in order to work out whether it is trying to access the hotspot for longer than the permitted time.
- **Tracking a device:** Some companies and organisations track devices (and therefore their users) by checking which wireless access points have been accessed by specific MAC addresses.
- **Assigning 'static' or 'fixed' IP addresses:** Each time a device connects to a network, it is identified by a DHCP server (usually using its MAC) and given the same IP address as before.

Vocabulary

- **Universally unique:** The only one of its kind in the world.
- **Identifier:** A group of letters, numbers, or symbols that a computer has been programmed to recognise and use to process information.
- **Hotspot:** A place in a public building where there is a computer system with an access point, which allows people in the building with a wireless computer or Bluetooth mobile phone to connect to a service such as the internet.

Device Name

A device name is a descriptive name that helps users to identify computers on a network. Device names are not used by computers to communicate with each other as they are not always unique. This means that they could cause conflicts if data was sent to more than one device with the same name for processing. You can change a device name using tools in the device's operating system.

COMPONENTS OF WIRED AND WIRELESS SYSTEMS

Cables

- **Cat5e cables:** Used in homes and small businesses for Ethernet connections. Able to transfer data at 10 Mbit/s, 100 Mbit/s or 1 Gbit/s. Connects devices through their NICs.
- **Cat6 cables:** Used to transfer data at 10 Gbit/s. These are more expensive than Cat5e cables and are usually only used by businesses.
- **Fibre optic cables:** Flexible fibres containing a glass thread that bounces light signals between two devices faster and further than is possible with wire cables. Can carry data at 40 Gbit/s over many kilometres without affecting signal quality. They are expensive and used by telecommunications companies, science/engineering laboratories, hospitals, banks, schools and universities.

Wireless Access Point (WAP)

A wireless access point allows devices with Wi-Fi connectivity to connect to a wired network. They are often built into other hardware, such as routers, but they are also available as stand-alone devices that connect to a wired network using Ethernet cables.

Switch

A switch connects devices on a network. It has ports, each of which can be connected to a device using a cable. Connecting a wireless access point to a switch via a cable gives wireless devices access to the wired network. The switch makes sure that data sent from any device gets to the correct device on the network.

Gateway

A gateway connects two different types of network. For example, a LAN is connected to a WAN using a gateway.

Router

A router stores the addresses of all devices that are connected to it so that it can forward network traffic to its destination using the quickest route. Most routers used in homes include

a switch and a wireless access point. They also act as gateways, connecting the LAN to the internet, which is a type of WAN. Home routers are dynamically allocated an IP address by the internet service provider (ISP).

Booster

A booster is used to amplify the signal in a network so that its range can be extended. For homes and offices, wired Ethernet connections often have a maximum range of 100 m. Wireless signals have limited range, too. Boosters can be used for both wired and wireless connections. Wireless access points can be set to repeater mode in order to act as boosters for Wi-Fi signals.

SERVERS

A server is a computer that shares its resources with connected devices. Computers connected to a server are known as clients. A single computer can be used to perform multiple server functions, though performing multiple roles will affect its performance.

Server Types & Functions

- **Authentication Server:** Checks usernames and passwords. When a user successfully logs in, the client receives an electronic certificate that it can then use to access various resources, including applications and storage.
- **Print Server:** Manages multiple printers at a time, dealing with print requests from client computers and adding jobs to a queue so that individual printers are not overloaded with requests. They can also monitor and process print requests to invoice users or departments.
- **File Server:** Allows users to access shared and private storage.
- **Application Server:** Provides clients with access to applications that can be run directly from the server.
- **Web Server:** Processes requests for data made via Hypertext Transfer Protocol (HTTP). Together, all of the content stored on all web servers is known as the World Wide Web

Vocabulary

- **Client:** A computer connected to a server.
- **Hypertext Transfer Protocol (HTTP):** A set of standards that control how computer documents that are written in HTML connect to each other.

CONNECTING TO AND USING THE INTERNET

Internet Service Provider (ISP)

To connect to the internet, users need to subscribe to an ISP. ISPs provide connections to the telecommunications infrastructure that forms the framework for the internet. ISPs provide access via mobile telephone networks and landline telephone networks. Commercial ISPs charge subscription fees for access to the internet. Some ISPs provide free access as part of community schemes.

Web Browser

A web browser is a type of software application used to request and display information stored on web servers. Examples of web browsers are Mozilla Firefox, Google Chrome, Internet Explorer or Microsoft Edge, Opera, and Safari.

Search Engine

A search engine provides users with a way to find information in web pages stored on web servers. Users enter keywords that describe the information they want to find. The search engine then compares the keywords with those in its database of web pages and returns the results that are the closest match to the given keywords.

Hint: Many people confuse search engines with web browsers. A search engine allows users to search and find information using keywords. A web browser is an application that users can use to view online content. Users can use a web browser to access a search engine.

Filter Software

Filter software prevents users from accessing inappropriate information. When a user tries to access a web page, the address (URL) and/or the contents of the web page are compared against two lists of URLs and keywords stored in the filter software's database: the blacklist and the whitelist.

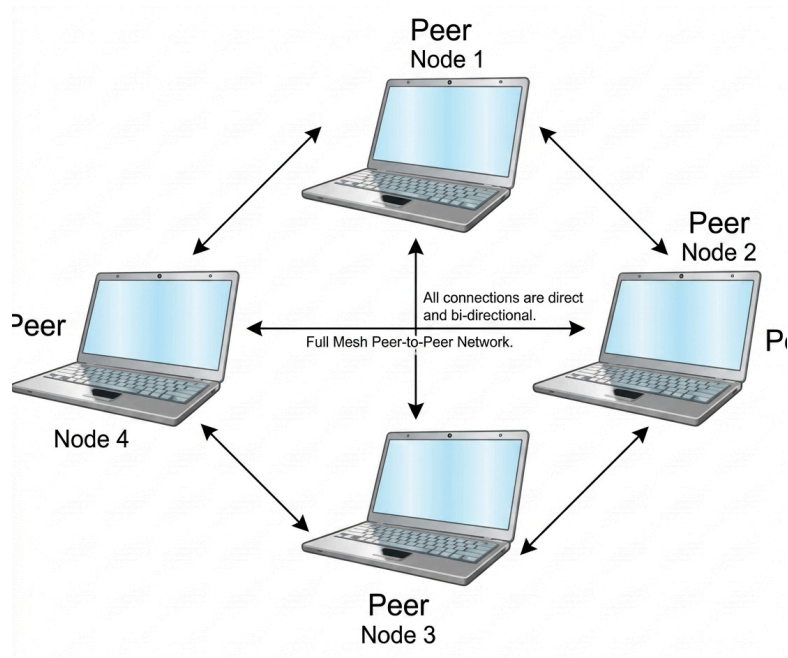
- If the results match anything in the **blacklist**, the user will be prevented from viewing the web page.
- If the result matches anything in the **whitelist**, then the user will be allowed to view the web page.
- If the result does not match anything in either list, the user will be allowed to view the information.
 - **URL (uniform resource locator):** A website address.
 - **Blacklist:** A list of unacceptable URLs.
 - **Whitelist:** A list of acceptable URLs.

LOCAL AREA NETWORKS (LANs)

A LAN is a network contained to a small area, such as a home or office network. Computers in a network can be connected using one of two different models:

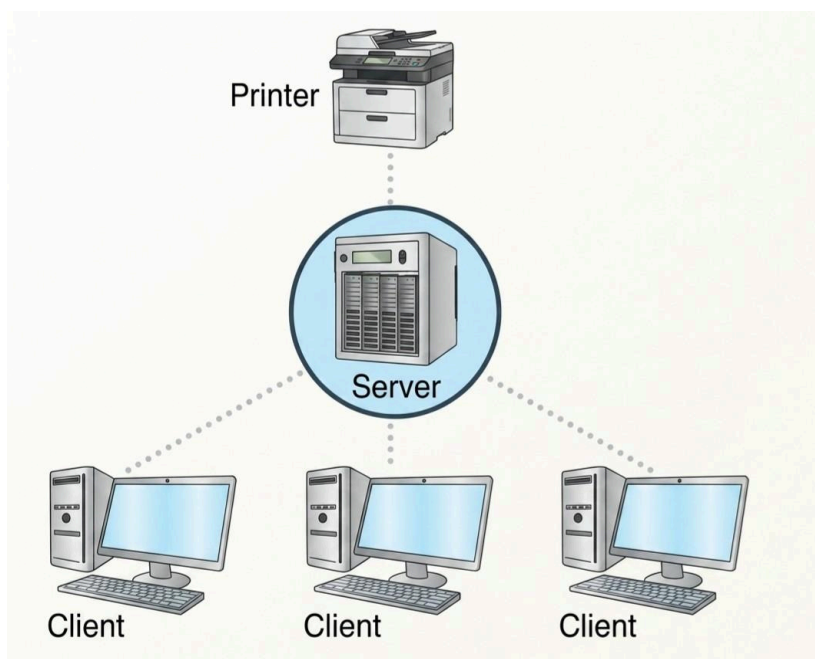
1. Peer-to-Peer Networks

Computers in a peer-to-peer network share their resources with other computers in the network, but they do not access servers.



2. Client-Server Networks

Some networks use servers. A network that uses servers and clients is called a client-server network.



Benefits of Using LANs

- Access to shared peripherals
- Access to shared storage and data
- Flexible access (being able to access peripherals, storage and data from any connected device)
- Media streaming (including movies, music and gaming)
- Communication (being able to send messages and files to others on the network)
- Shared access to the internet

Benefits of Using Client-Server Networks

- **Control of user access rights:** Users, or groups of users, can be given access to some resources (such as storage or printers) and restricted from accessing others.
- **Centralised administration:** Resources and user accounts can be managed by an individual, or individual group of servers and administrators. This ensures support is provided by people who have an overview of the network.
- **Centralised backup:** User data is protected from loss because backups can be automated for all users.
- **Shared software:** Application servers can provide access to shared software or operating systems.
- **Shared storage and file access:** Storage amounts can be managed centrally, allowing users to make files available to others with specific file permissions (read-only or read-and-write).
- **Roaming profiles:** This is the ability to log into any computer in an office and see your settings and files. This enables users to work from anywhere there is a client.

SECURING DATA ON NETWORKS

Security prevents unauthorised users from accessing network resources and data.

Logins and Passwords

Users log in to computers on a network to access centrally managed resources. Passwords are used to authenticate a user to the network. An authentication server can manage the authentication across a range of services so a user does not have to log in each time they access a resource.

- **Login:** User or account information, such as a user name or account name.
- **Authenticate:** Confirm that the user is who they say they are.

Firewalls

A firewall is used at the gateway to a network. It controls the network traffic to and from a network, particularly the traffic from the internet. Firewalls prevent unauthorised users from accessing network devices and resources, such as storage. Firewalls are available as hardware and software.

Encryption

Encryption is the process of encoding, scrambling or jumbling data so that unauthorised users are prevented from being able to understand it.

- **Symmetric Key Encryption:** Uses the same key at both ends of the process, meaning that the same key is used to encrypt and decrypt the data.
- **Public Key Encryption:** Uses two mathematically related keys called a key pair. One key (public key) is shared with other computers to encrypt the data, and a different, unshared key (private key) is used by the recipient to decrypt it.

Vocabulary

- **Cipher:** A code.

Wireless Protocols: WEP vs. WPA

It is easier to intercept data in a wireless network than in a wired network.

- **Wireless Encryption Protocol (WEP):** Used to secure the wireless transfer of data. It is the least secure wireless data encryption method because every device on the wireless network uses the same key for every transfer. Eavesdroppers studying enough packets can easily identify the key.
- **Wi-Fi Protected Access (WPA):** A security protocol designed to provide better encryption than WEP. WPA generates a new key for each device on the wireless network, and new keys are provided for each packet of data sent.
 - **Eavesdropper:** An unauthorised person or piece of software that intercepts data from a private communication.
 - **Packet:** A unit of data packaged to travel across a network.

Virtual Private Network (VPN)

A VPN provides access to a private LAN from a remote location. The connection to the LAN is created using the infrastructure of a public network like the internet. Data sent using a VPN is encrypted so that it is secure if it is intercepted. An individual might use a VPN to:

- Access their employer's network when working from home.
- Access computers in a different geographical location, perhaps to avoid local restrictions on access to web content (such as due to censorship or geolocation rights management).
- Make secure payments.
- Prevent surveillance of and access to their web activity.

Vocabulary

- **Geolocation rights management:** The use of a user's location to block access to services online.

File Access Rights, Logs, and Backups

- **File Access Rights:** Also known as file permissions. They can be set for individual files, folders or drives, and they ensure that users are either allowed to read only or allowed to read and write.
- **Transaction Logs:** All network activity can be recorded in a log file. A transaction log helps to identify which computers and network devices have been accessed, allowing administrators to identify unusual activity.
- **Backups:** A backup is a copy of one or more files. The backup or backups are usually stored on a different storage device to the original file.

Vocabulary

- **Log:** A record of events.
- **Transaction:** The exchange of data.

CHAPTER QUESTIONS

1. Which one of these connects a LAN to a WAN? (1)
 - A: Switch
 - B: Gateway
 - C: Modem
 - D: Server
2. Explain why public key encryption is more secure than symmetric key encryption. (2)
3. State which security device controls the traffic entering a network. (1)
4. State two methods by which devices are identified to each other on a network. (2)
5. Explain why IPv6 was used to replace IPv4. (2)
6. State two uses for a VPN. (2)
7. State the reason why a booster is used in a network. (1)
8. Describe how encryption secures data. (3)
9. List three methods of securing data on networks. (3)
10. State three benefits to users of using a local area network. (3)
11. Draw and label a network diagram for a home that includes: (8)
 - Two smartphones
 - One tablet device
 - Two desktop PCs
 - Internet access